Student Number:									

Surname:																
First Name(s):																

Department of Mathematics and Applied Mathematics

MAM2000W: 2LA (Linear Algebra)

Test 2 **3 May 2018**

Time : 2 hours **Full marks: 60**

- This question paper consists of 8 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. The work in your *UCT Examination Answer Book* will not be marked. Only the answers that you write on this question paper will be marked.
- Calculators are not allowed. When your answer contains constants such as π, e or $\sqrt{2}$, leave them in that form. Don't simplify your answers.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	TOTAL
7	8	9	9	12	6	9	60

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Question 1. [7 points]

a. Write down the elementary matrix for replacing row 3 by itself plus 4 times row 1 in a 3×3 matrix.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{pmatrix}.$$

b. Suppose A is a matrix for which $\det(A) = 3$ and $\det(2A) = 12$. Then

$\det(A^T) = 3, \qquad \det(A^{-1}) = 1/3,$

and the dimensions of A are (3)

$2 \times 2.$

c. Use Cramer's rule to find y if

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

(Show what formula you are using.)

$$y = \frac{\det \begin{pmatrix} 1 & 5 \\ 3 & 6 \end{pmatrix}}{\det \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}} = \frac{-9}{-2} = 4.5.$$

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Question 2. **[8 points]**

a. Use elementary row operations to find the determinant of

$$A = \begin{pmatrix} 1 & 5 & -6 \\ -1 & -4 & 4 \\ -2 & -7 & 9 \end{pmatrix}.$$

Write down what operations you are doing! (4)

$$\begin{vmatrix} 1 & 5 & -6 \\ -1 & -4 & 4 \\ -2 & -7 & 9 \end{vmatrix} =^{R_2 \leftarrow R_2 + R_1 \text{ and } R_3 \leftarrow R_3 + 2R_1} \begin{vmatrix} 1 & 5 & -6 \\ 0 & 1 & -2 \\ 0 & 3 & -3 \end{vmatrix}$$

$$=^{R_3 \leftarrow R_3 - 3R_2} \begin{vmatrix} 1 & 5 & -6 \\ 0 & 1 & -2 \\ 0 & 0 & 3 \end{vmatrix}$$

$$= 3$$

b. How many solutions does the equation $Ax = 0$ have for x? Why? (2)

One, because $\det(A) \neq 0$ so A is invertible.

c. Are the columns of A linearly independent? Why, or why not? (2)

Yes, because $Ax = 0$ has only the trivial solution $x = 0$.

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Question 3. [9 points]

Let A and B be $n \times n$ matrices with $AB = I$. We want to prove that then $BA = I$ also.

a. Show that the equation $Ax = b$ has a solution for x for every $b \in V_n$.

Let $x = Bb$. Then

$$Ax = A(Bb) = (AB)b = b = b,$$

so x is a solution to $Ax = b$.

b. Why does this imply that the row reduced echelon form of A has no zero row?

Let A' be the reduced echelon form of A, and note that for every $c \in V_m$ we can, by reversing the steps of our Gauss reduction, find a vector b such that $(A|b) \sim (A'|c)$. If A' has a zero row, then choosing any vector c whose last entry is non-zero results in a vector b for which $Ax = b$ has no solution.

c. Why does this imply that A can be row reduced to the identity matrix I?

If A' has no zero row then every row has a pivot. But A is an $n \times n$ matrix, so this implies every column is a pivot column, and hence the reduced row echelon form is I_n.

d. Complete the proof that $BA = I$. (You can use the statements in A., b, and c., even if you haven't proved them.)

Since $A \sim I$, there exist elementary matrices $E_1, \ldots, E_p$ such that

$$(E_p \cdots E_2 E_1)A = I.$$

So (multiplying both sides on the right by B and using associativity) we have

$$E_p, \ldots, E_1(AB) = IB.$$

But $AB = I$, so $E_p, \ldots, E_1 = B$, and hence $BA = I$.

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Question 4. **[9 points]**

a. Let V be the vector space of all real-valued functions of a real variable. Show that the set

$$W = \{f \in V \mid f(5) = f(3)\}$$

is a subspace of V, justifying each step in your proof. (5)

W contains the zero polynomial, and so it is not empty. Now let $f, g \in W$ and $a, b \in \mathbb{R}$. Then

$$\begin{aligned}(af + bg)(5) &= af(5) + bg(5) \\ &= af(3) + bg(3) \text{ since } f, g \in W \\ &= (af + bg)(3),\end{aligned}$$

so $af + bg \in W$. Hence W is closed under linear combinations, and is a subspace.

b. Let P be the subset of V consisting of polynomials of degree exactly 5. Show that P is not a subspace. (2)

P does not contain the zero vector. OR P is not closed under addition, since for example $f(x) = x^5 - 1 \in P$ and $g(x) = -x^5 + 2x \in P$, but $(f + g)(x) = 2x - 1 \notin P$.

c. In $\mathbb{R}^3$, describe geometrically the subspace $\langle(1, 2, 3)\rangle$. (2)

It is a straight line through the origin.

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Question 5. [12 points]

a. Find the inverse of $A = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 2 & 3 & 4 \end{pmatrix}$, or show that A is not invertible. Write down the operations you are doing! (4)

$$A^{-1} = \begin{pmatrix} 2 & 5 & -3 \\ 0 & -2 & 1 \\ -1 & -1 & 1 \end{pmatrix}.$$

b. The set $\{1, x, x^2\}$ is a basis for the vector space $\mathbb{R}_2[x]$. Determine whether the set

$$S = \{1 + x + 2x^2, 2 + x + 3x^2, 1 + 2x + 4x^2\}$$

is linearly independent or dependent. (6)

(Hint: You might want to use your answer from a. to save time.)

Consider a linear relation on S:

$$a(1 + x + 2x^2) + b(2 + x + 3x^2) + c(1 + 2x + 4x^2) = 0 \qquad (1)$$

for some $a, b, c \in \mathbb{R}$ then

$$(a + 2b + c) + (a + b + 2c)x + (2a + 3b + 4c)x^2 = 0 \text{ for all } x \in \mathbb{R},$$

which (since $\{1, x, x^2\}$ is linearly independent) is true if and only if

$$\begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Since this matrix is invertible (proved in (a)) this has only the trivial solution $a = b = c = 0$, and it follows that S is linearly independent.

c. Is S a spanning set for $\mathbb{R}_2[x]$? Why, or why not? (2)

Yes, because S contains three vectors and $\mathbb{R}_2[x]$ has dimension 3.

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Question 6. [6 points] Prove **one** of the following two theorems:

a. If A is a subset of a vector space V, then $\langle A \rangle$ is equal to the intersection of all the subspaces of V that contain the set A.

Let W be any subspace of V that contains A. Then since W is closed under linear combinations, $\langle A \rangle$ is contained in W. It follows that $\langle A \rangle$ is contained in every such subspace W, and hence

$$\langle A \rangle \subseteq \bigcap \{W \leq V | A \subseteq W\}.$$

On the other hand, since $\langle A \rangle$ is itself a subspace containing A, it must contain the intersection of all such subspaces. Thus

$$\bigcap \{W \leq V | A \subseteq W\} \subseteq \langle A \rangle,$$

and so the sets are equal.

b. If A and B are subspaces of a vector space V and $A \cup B$ is a subspace of V, then one of A and B contains the other.

If $W_1 \subseteq W_2$ then $W_1 \cup W_2 = W_2$, which is a subspace. If $W_2 \subseteq W_1$ then $W_1 \cup W_2 = W_1$, which is a subspace.

Now let $W_1 \not\subseteq W_2$ and $W_2 \not\subseteq W_1$. Then there exists $w_1 \in W_1$ such that $w_1 \notin W_2$ and $w_2 \in W_2$ such that $w_2 \notin W_1$. However, we claim that neither $w_1 + w_2 \in W_1$ nor $w_1 + w_2 \in W_2$, so we conclude $w_1 + w_2 \notin W_1 \cup W_2$, and hence $W_1 \cup W_2$ cannot be a subspace (because it is not closed under vector addition). In order to prove the claim, first suppose that $w_1 + w_2 \in W_1$. Then, since $w_1 \in W_1$ and $-w_1 \in W_1$ we must have

$$w_2 = (w_1 + w_2) - w_1 \in W_1,$$

which is not true. We have just derived a contradiction, so we cannot have $w_1 + w_2 \in W_1$. Similarly, if $w_1 + w_2 \in W_2$ then

$$w_1 = (w_1 + w_2) - w_2 \in W_2,$$

which is also not true. Therefore we conclude that $w_1 + w_2$ cannot be in either W_1 or in W_2, so $w_1 + w_2 \notin W_1 \cup W_2$.

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Question 7. [**9 points**]

a. State the Steinitz Replacement Theorem, without proof. (4)

Let $A = \{\alpha_1, \ldots, \alpha_m\}$ and $B = \{\beta_1 \ldots, \beta_n\}$ be subsets of a vector space V such that A is linearly independent and B spans A. Then

(i) $m \leq n$ (i.e, A cannot be larger than B) and,

(ii) There are some m vectors in B which can be replaced by the m vectors in A without changing the span. That is, we can find a way to reorder the elements of B so that

$$\langle\{\alpha_1, \ldots, \alpha_m, \beta_{m+1}, \ldots, \beta_n\}\rangle = \langle B\rangle.$$

b. Prove that every linearly independent set of vectors in an n-dimensional vector space V can be extended to a basis for V. (5)

Let A be a linearly independent set in V, and let B be a basis for V. Then (since A is linearly independent and B is a finite spanning set for V and hence A), by the Steinitz Replacement Theorem we can replace $|A| \leq n$ of the vectors in B by the $|A|$ vectors in A to get a new set B' which also spans V.
But $|B'| \leq |B| = n$, so in fact $|B'| = n$ (it cannot be smaller than n because it is a spanning set) and by the Basis Theorem B' is a basis for V. Since B' contains A, we are done.

OR

Let A be a linearly independent set of vectors in V. Then by the Basis theorem $|A| \leq n$. We want to show that A can be extended to a basis.
If A spans V, then it is a basis and we are done. Otherwise there is some vector $\alpha \in V - \langle A\rangle$ (a vector α in V that cannot be written as a linear combination of vectors in A). But then $A \cup \{\alpha\}$ is a linearly independent set too.
If this is a spanning set of V, then it is a basis containing A, and we are done. Otherwise, we repeat the process, adding vectors to our set while keeping it linearly independent, until we reach a spanning set. This process will terminate when our linearly independent set reaches size n, because by the Basis theorem it must then be a basis. Since it contains A, we are done.

c. **(Bonus question)**

Hence prove that if W is a subspace of an n-dimensional vector space V then there exists a subspace U of V such that $V = W \oplus U$. (4)

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Suppose $\dim(W) = k \le n$, and let $B_W = \{\beta_1, \dots, \beta_k\}$ be a basis for W. Since B_W is linearly independent, it can be extended to a basis

$$B_V = \{\beta_1, \dots, \beta_k, \beta_{k+1}, \dots, \beta_n\}$$

for V. Now let $U = \langle \beta_{k+1}, \dots, \beta_n \rangle$.
We first show that $V = W + U$. Since $\langle B_V \rangle = V$, every vector $\alpha \in V$ can be written as

$$\alpha = \sum_{i=1}^{n} a_i\beta_i = \underbrace{\sum_{i=1}^{k} a_i\beta_i}_{\in W} + \underbrace{\sum_{i=k+1}^{n} a_i\beta_i}_{\in U} \in W + U$$

for some scalars $a_1, \dots, a_n$. So $V \subseteq W + U$. But $W + U \subseteq V$ since V is closed under addition, so $V = W + U$.
We now show that $W \cap U = \{0\}$. Suppose $\gamma \in W \cap U$. Then

$$\gamma = \sum_{i=1}^{k} c_i\beta_i \text{ and } \gamma = \sum_{i=k+1}^{n} c_i\beta_i$$

for some scalars $c_1, \dots, c_n$. But then

$$\sum_{i=1}^{k} c_i\beta_i - \sum_{i=k+1}^{n} c_i\beta_i = 0$$

is a linear relation on B_V, and since B_V is linearly independent it follows that all the c_i are zero, i.e., that $\gamma = 0$ and $W \cap U = \{0\}$.